



Management System Research and Implementation Strategy

System Implementation and Process Optimization
- A Data-Driven Implementation Journey -



Current State Assessment

Background



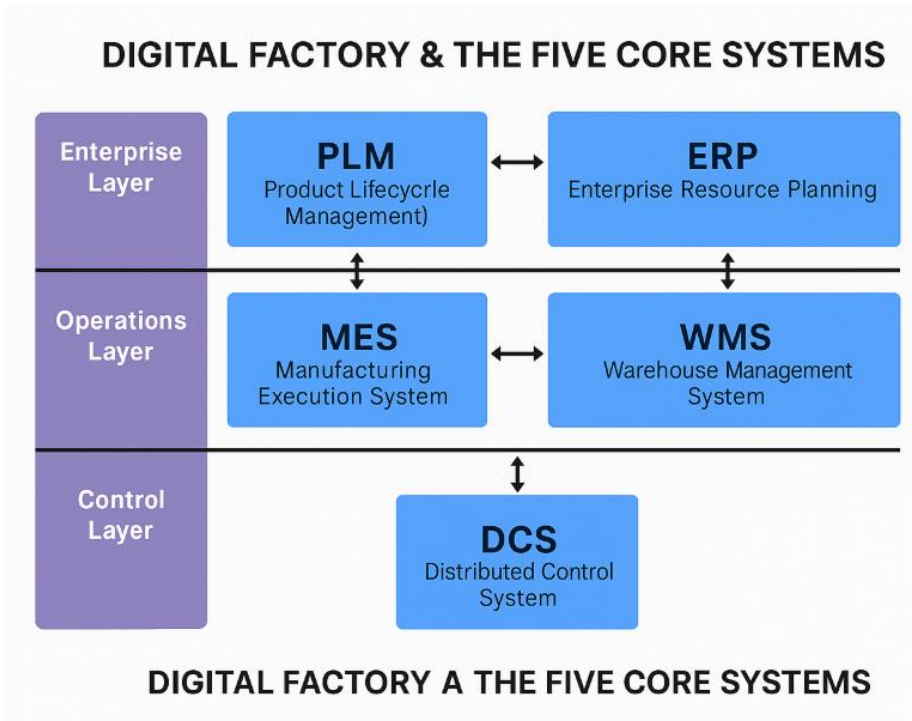
- ☐ The current systems are insufficient to support material inventory management across the factory
- ☐ Raw materials, semi-finished goods, finished goods, and accessories are managed manually
- ☐ Receiving, storage, inventory tracking, material movement, and shipping rely on manual operations
- ☐ This results in limited inventory visibility, inconsistent processes, and reduced inventory accuracy

Purpose



- ☐ Assess current warehouse operations and identify operational gaps
- ☐ Benchmark leading system solutions and industry best practices
- ☐ Evaluate feasible implementation approaches
- ☐ Recommend the most suitable solution for warehouse optimization

Digital Factory Systems Concept



Enterprise Layer

PLM – Product Lifecycle Management

- Drawings | Sizes | Materials | Bill of Materials

ERP – Enterprise Resource Planning

- Planning and resource allocation

Operations Layer

MES – Manufacturing Execution System

- Projects -> Orders -> Work Orders: 1, 2, 3 ...

WMS - Warehouse Management System

- Inbound/outbound storage, location and allocation of raw materials, semi-finished products and finished products

WMS bridges warehouse operations with production and enterprise planning

Functional Requirements

Functional

Inventory Management

- Inbound
- Outbound
- Allocation
- Inventory Count
- Barcode

Multi-Languages

- English
- Chinese
- Spanish

Low Cost

Optional

Inventory Management

- Multi-locations management
- BOM components management
- Logistics tracking

Purchase/Sales Orders Management

- Vendor: Demand -> Purchase Order (Inbound)
- Customer: Quote -> Sales Order (Outbound)

Production Management

- Work Orders
- Replacement Material Management

Software integration and unification

- Collaboration with Excel (operational processes)
- Integrating or replacing QuickBooks

System Solution Comparison

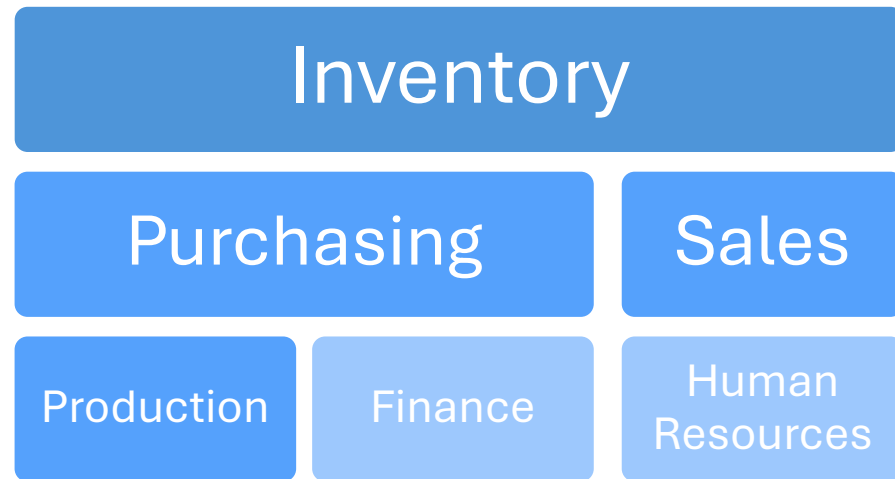
ERP	Inventory / Manufacturing System	Low-Code Platform
American: SAP, Netsuite, Epicor, Dynamics 365, Odoo, ERPNext Chinese: Kingdee, Yonyou	American: Inflow inventory, SOS Inventory, Quickbooks Chinese: Kingdee Cloud Galaxy	American: Artable Chinese: Jiandaoyun, Yunbiao

Solution	Type	Key Strengths	Key Limitations	Estimated Cost
ERPNext	ERP	Inventory, purchasing, sales and manufacturing; customizable	Requires configuration and implementation support	Low
Odoo, Quickbooks Desktop	ERP	Comprehensive modules, scalable and strong integration capability	Higher cost and implementation complexity	Medium–High
Kingdee Ailit	Inventory System	Chinese-language support, inventory management and mobile access	Limited manufacturing and U.S. system integration	Low
Inflow Inventory	Inventory System	Easy to use, strong inventory and order management	Limited multilingual support and software integration	Medium

ERPNext provides the best balance of functionality, flexibility, and cost for current business needs

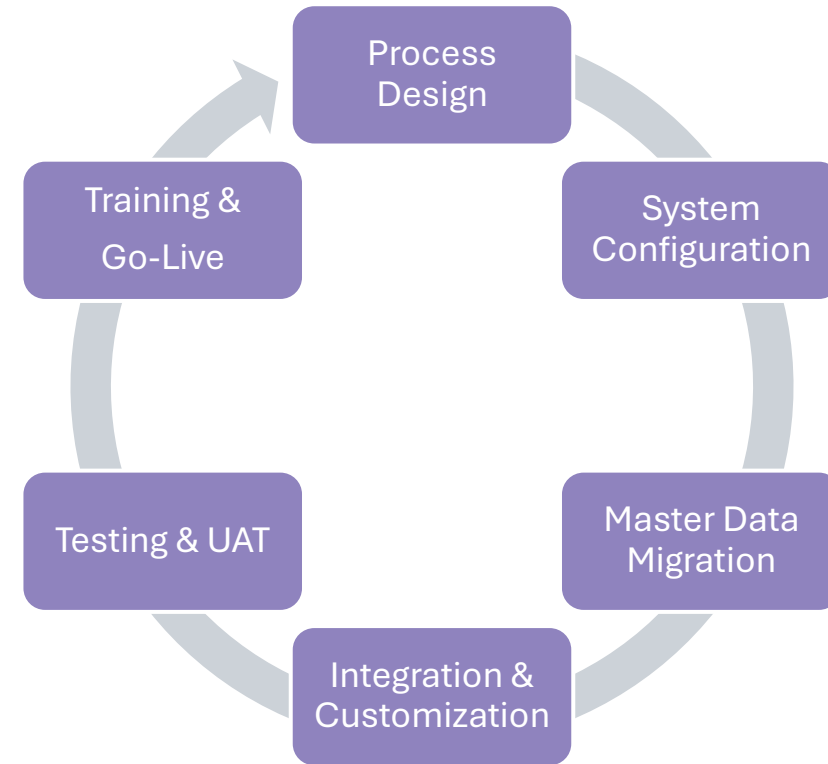
System Implementation Roadmap

Core Modules



Inventory Management is the highest priority for Phase 1

Implementation phases



A phased implementation approach is recommended to reduce deployment risk

ERPNext Deployment

Testing: 1-2 users

Help

\$5 /mo
\$0.16/day

0.5 compute hours / day
250 MB Database
2.5 GB Disk

\$10 /mo
\$0.32/day

1 compute hour / day
500 MB Database
5 GB Disk

\$25 /mo
\$0.81/day

2 compute hours / day
1 GB Database
25 GB Disk
Offsite Backups

\$38 /mo
\$1.21/day

3 compute hours / day
1.5 GB Database
37.5 GB Disk
Offsite Backups

\$50 /mo ✓
\$1.61/day

4 compute hours / day
2 GB Database
50 GB Disk
Product Warranty
Database Access
Offsite Backups
Advanced Monitoring

\$75 /mo ✓
\$2.42/day

6 compute hours / day
3 GB Database
75 GB Disk
Product Warranty
Database Access
Offsite Backups
Advanced Monitoring

\$100 /mo
\$3.23/day

8 compute hours / day
4 GB Database
100 GB Disk
Product Warranty
Database Access
Offsite Backups
Advanced Monitoring

\$150 /mo
\$4.70/day

12 compute hours / day
6 GB Database
150 GB Disk
Product Warranty
Database Access
Offsite Backups
Advanced Monitoring

\$200 /mo
\$6.46/day

16 compute hours / day
8 GB Database
200 GB Disk
Product Warranty
Database Access
Offsite Backups
Advanced Monitoring

Using: up to 10 users, only \$5 per user

ERPNext Opensource Documents



frappe / erpnext



Code

Issues

1.8k

Pull requests

68

Actions

Projects

Wiki

Security and quality

14



erpnext

Public



Watch

675



Fork



develop



Go to file



Code



mihir-kandoi Merge pull request #57223 from aerele...



a311193

6 minutes ago



.github

docs: catalog second-order GROUP B...

2 weeks ago

.greptile

docs: catalog second-order GROUP B...

2 weeks ago

banking

chore: update dependencies in banki...

2 weeks ago

erpnext

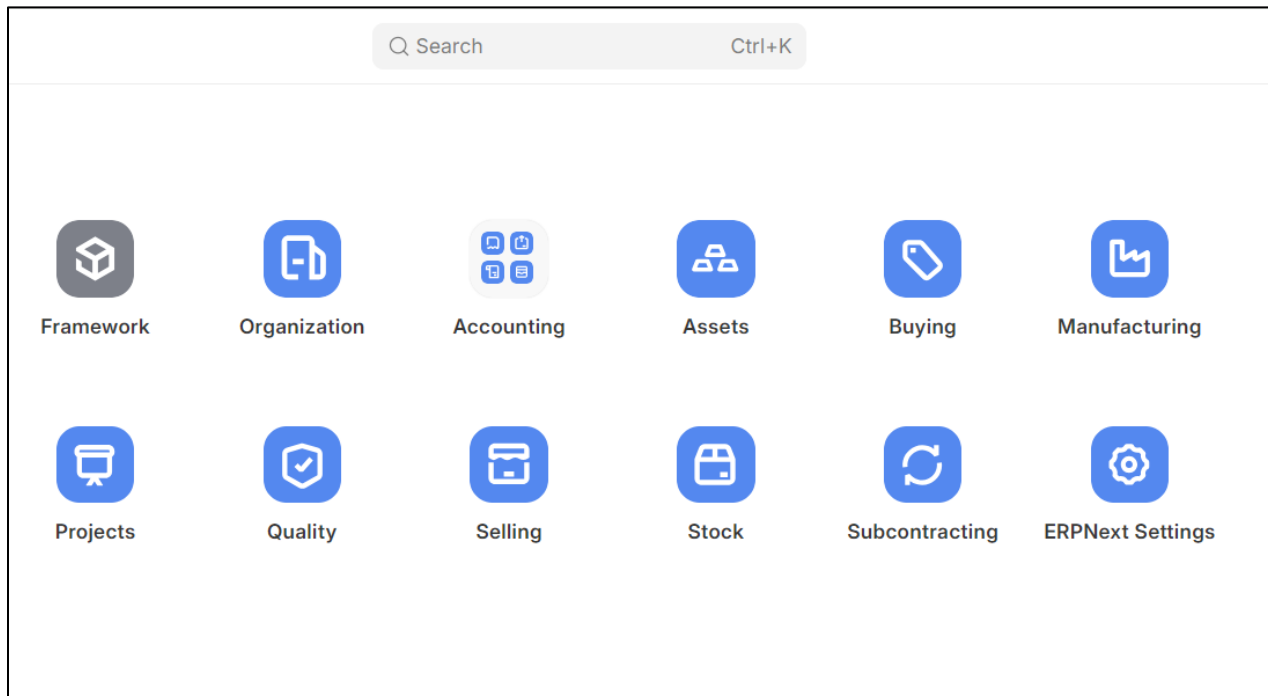
Merge pull request #57223 from aer...

6 minutes ago

ERPNext Demonstration

System Link

<https://shop.cfacabinetry.store/desk>



Marketing Link

<https://shop.cfacabinetry.store/Products>

